

Reverse Auctions: Driving India's Renewable Energy Expansion

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1. Introduction

Over 80% of India's energy needs have been met by energy generated through coal, oil and solid biomass. However, this does not fulfil the energy demands of India's growing population and by relation GDP, which has grown exponentially since 2000. (International Energy Agency, 2021) The Indian Government turned to renewable energy (RE) to address the need and has incentivised production through schemes like the Jawaharlal Nehru National Solar Mission (JNNSM), which aimed to increase solar capacity installation to 100 GW by 2022. (Ministry of New and Renewable Energy (MNRE), 2025).

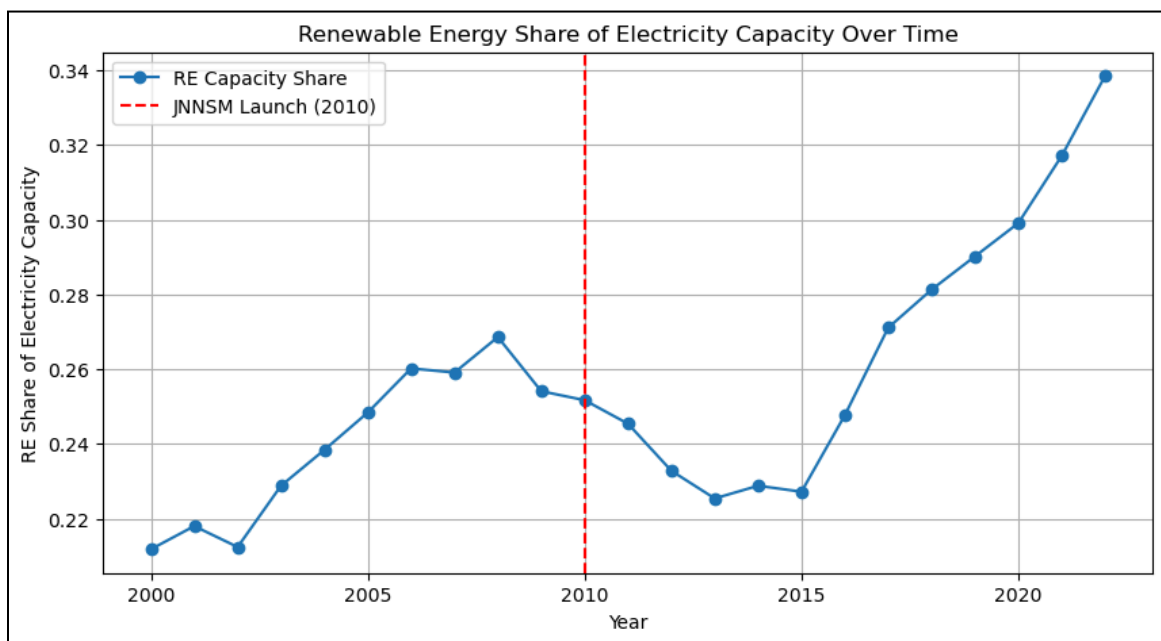


Figure 1: Renewable Energy Share of Electricity Capacity Over Time

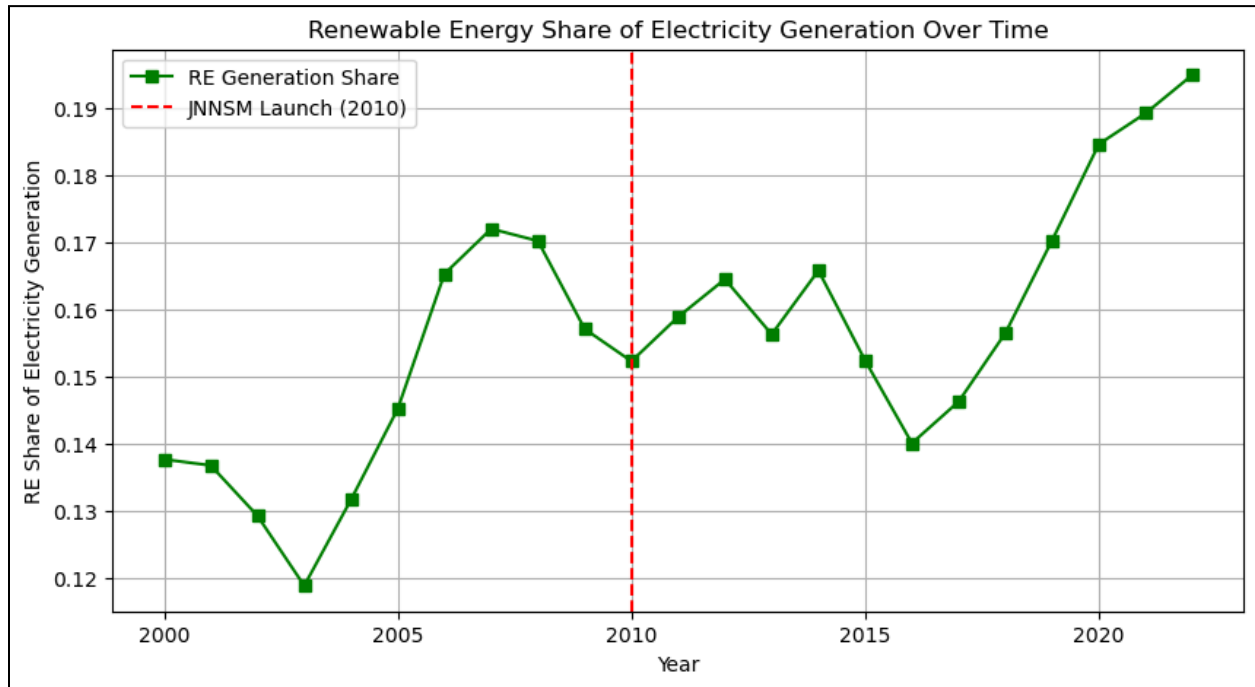


Figure 2: Renewable Energy Share of Electricity Generation Over Time

Source: International Renewable Energy Agency

The renewable energy share in India's electricity capacity increased after 2015, reaching 34% by 2022. India is now home to some of the largest solar parks in the world, and its solar power capacity lies at 102.57 GW as of February 2025. This aligns with the launch of solar policies. However, the RE generation share as a source of energy has grown less consistently. This may result from capacity utilization problems, integration challenges, and feasibility issues. While capacity expansion was rapid, actual generation was constrained by transmission delays and market dynamics. The post-2015 upward trend suggests improvements in policy, grid stability, and technology, but it also implies that ensuring efficient utilization of renewable energy remains a challenge.

Behind these trends in the RE Sector is a series of policies incentivizing solar power projects by subsidizing their financing. The Government of India (GoI) has undertaken various strategies, including the internationally used Feed-in Tariffs (FIT) where regulators set fixed prices for renewable energy, with the government covering the cost difference, creating a fiscal burden. However, GoI plans to phase out these tariffs and introduce e-Reverse Auctions (e-RA) for procurement of production contracts. With these auctions, bidders submit offers and compete

to lower tariffs. This market-driven approach has reduced government subsidies and made power procurement more cost-effective, replacing the FIT system.

The auction is called e-RA because it is conducted electronically. While reverse bids may be a cost-effective policy that increases price competition and reduces subsidies, they resulted in considerable technical failures in solar panel installations, which may deter foreign investors in the future (Münch & Marian, 2022).

Our paper examines data from government press releases and research reports to gauge the reverse auctions' effectiveness in increasing RE capacity and decreasing electricity prices while maintaining the projects' financial viability. We will discuss allocation measures, price reduction, as well as comparing bids before and after the introduction of the e-RA. We also break down the auction format and delve into scenarios where producers might experience a winner's curse, highlighting any inefficiencies posed through this procurement method.

2. Literature review

The literature focusing on the reverse auction mechanism as a tool for increasing renewable energy capacity in India is quite limited. However, existing literature provides some insights to the effectiveness of the bidding process internationally and, to a lesser extent, in India. Chao and Wilson (2002) designed a multidimensional procurement auction for solar power reserves in California, which resemble the e-RA. They assign a value function for the multidimensional capabilities of each bidder, which contributes the “score” assigned to each bid. The value function is discounted with the price bid to create a “score,” based on which the winning bid is declared. The score depicts the tradeoff between price and non-price attributes. Chao and Wilson's work shows the important role the scoring rules play in bidders' behavior and auction efficiency. This study is particularly important for designing optimal auction mechanisms and understanding the complex environment in the procurement process.

Maurer and Barroso (2011) conducted a cross-country analysis of auctions and concluded that although the bidding mechanism fosters a competitive environment and leads to lower prices, it faces two main risks: attracting bidders and ensuring project delivery. To address these risks, the authors recommend clear rules for the bidding process and pre-qualification requirements to filter out unreliable bidders - something that India has already introduced.

Additionally, Gephart et al. (2017) analyzed the European Union Member States that introduced a bidding system for renewable energy projects and concluded that while pre-qualification requirements and penalties enhance project realization rates, they also introduce higher risk premiums for bidders, which then drives price increase.

Focusing on India, Debnath, Mittal, and Jindal (2020) claim that the e-RA system was more cost-effective than the FIT but led to the winner's curse and, consequently, more failures to commission the promised capacity. The authors suggest that specific regulatory tools are needed to ensure the effectiveness of e-RA. For example, power purchase agreements (PPAs)¹ and renewable purchase obligations (RPOs)² would ensure the success of the commissioned projects by providing long-term energy contracts between the developer and the buyer and mandating large buyers to procure a certain percentage of their energy from renewable sources. On the other hand, such regulations carry a risk of higher electricity prices.

3. Auction Mechanism

Reverse auctions can be modeled as a strategic game in which bidders compete by undercutting one another's price bids. The lowest bid submitted is the winner, and the winning bidder pays the price they have quoted. This mechanism creates a competitive environment by making each developer strategically bid as near as their marginal cost of production. This ensures highly competitive pricing.

The e-RAs under discussion are organized using a Single-Stage Two-Envelope Bid format. Under this system, every bid the developer submits is multidimensional; it consists not just of the pricing bid but also includes an assessment of technical and financial capacity. This format resembles a scoring auction, where each bid is evaluated based on multiple criteria and assigned an overall "score." The winning bid is determined by which submission achieves the best (typically lowest) score according to the auctioneer's weighted evaluation. The auction design bids consist of two distinct components: technological and financial bid as well as pricing bid.

¹ Power Purchase Agreements are legally binding contracts between power producers and buyers, which define the terms and conditions for the purchase.

² Renewable Purchase Obligations are obligations that require large electricity consumers to purchase a specified quantity of renewable energy. They are used to encourage renewable energy adoption.

3.1. Technological and Financial Bid

This aspect reflects the bidder's overall strength and competency in managing both financial resources and technical operations. Bidders are expected to submit a Request for Selection (RfS) document which includes comprehensive details such as eligibility criteria, technical specifications and contractual conditions. The eligibility of a bidder is evaluated through the details provided in this document ([SECI, 2025](#)).

Adverse selection or the winner's curse in these auctions occur when bidders report their marginal costs of production incorrectly, in order to claim the winning bid. The policy tries to mitigate adverse selection by auditing the company's financial standing and technological capabilities.

3.2. Pricing Bid Composition

The pricing bid itself is divided into two key components:

- *Tariff*: The tariff is capped at Rs. 5.45 per kWh of energy produced, ensuring that bids remain within a predetermined competitive range.
- *Viability Gap Funding (VGF)*: The VGF mechanism allows producers to cover additional costs that go beyond the tariff. This funding is crucial for offsetting extraneous expenses and making the project financially feasible. However, the VGF is capped at 30% of the total project cost to maintain fiscal discipline and prevent excessive reliance on government support. Detailed guidelines on this mechanism are provided by SECI ([MNRE, 2015](#)).

The overall score $S(q,p)$ for each bid can be depicted as follows:

$$S(q, p) = V(q) - b_i$$

Where $V(q)$ is the value allotted to the quality of bid and b_i is the price bid. The value is a linear weighted function of quality parameters:

$$V(q) = w_1 f_i + w_2 t_i$$

Where f_i denotes the financial score and t_i denotes the technological capability.

The price bid b_i is constructed as follows:

$$b_i = p_i + \frac{VGF_i}{e_i}$$

where p_i is the tariff quote, VGF_i is the viability gap funding requested, p_i is the tariff quote.

This scoring function, adapted from the framework proposed by Chao and Wilson (2002), allows the auctioneer to balance cost competitiveness with the qualitative and quantitative strengths of each bid.

3.3. Short-Listing and Award Process

Initially, a short-listing process evaluates companies based on their technical and financial competency. Subsequently, only the pricing bids of these shortlisted bidders are considered. The winning bid is determined by calculating the production cost per kWh—specifically, the sum of the requested VGF divided by the production capacity the developer promises to fulfill and the tariff. Given the production capacity constraint, multiple bidders may be selected as winners to collectively satisfy the overall project demand.

3.4. Contractual Safeguards

To ensure compliance and accountability, the auction framework incorporates penalties and provisions for contract termination. These measures are designed to deter bidders from retracting their bids or failing to deliver on their production promises, thereby safeguarding the reliability of the awarded contracts.

4. Comparing Supply After e-RA Over FIT Systems

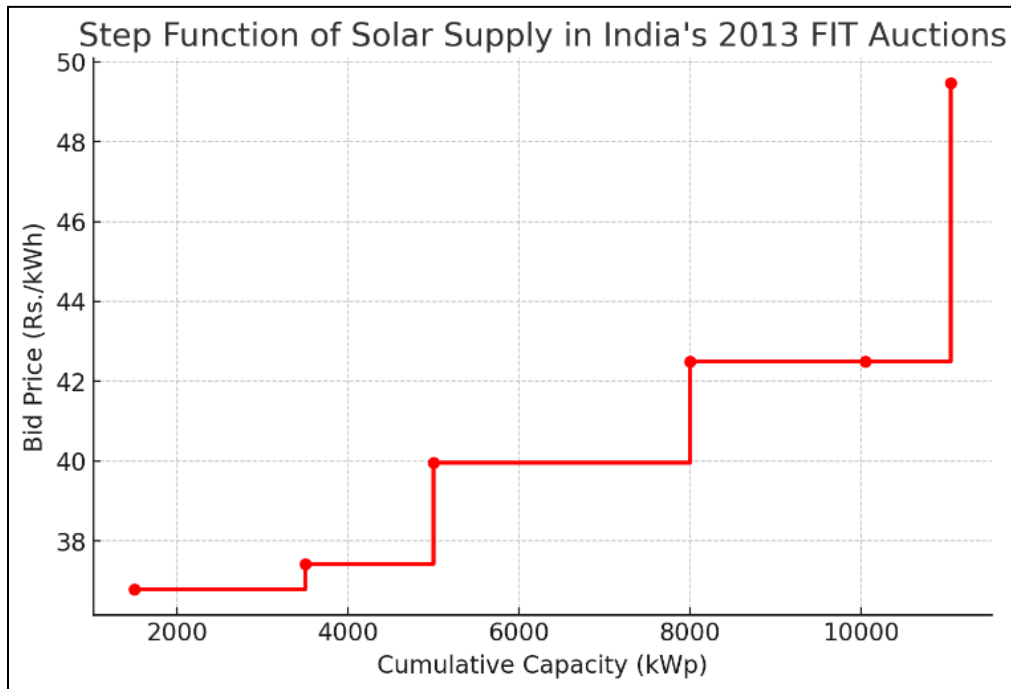


Figure 3: Step Function of Solar Supply in India's 2013 FIT Auctions

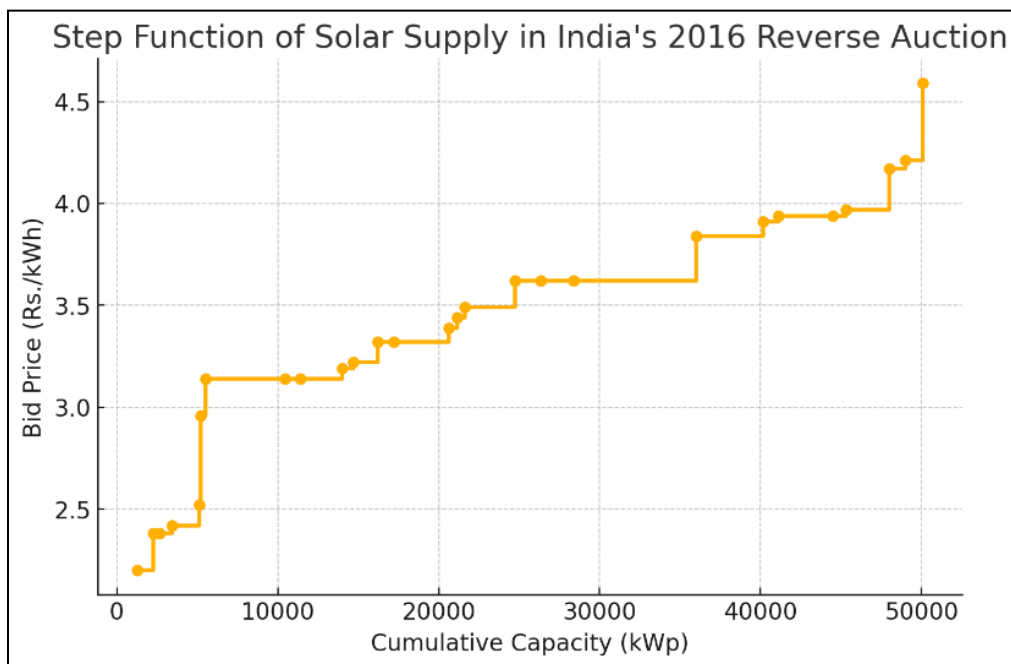


Figure 4: Step Function of Solar Supply in India's 2016 e-RA

Source: Solar Energy Corporation of India Limited

In 2013, India's solar energy procurement was primarily driven by the FIT system, where fixed prices were offered to incentivize solar development. We created the step function of supply for 2013 with the official data from Solar Energy Corporation of India Limited. Figure 3 demonstrates a steep curve, which indicates that additional capacity was only supplied at higher prices, ranging from ~35 to 50 Rs./kWh. This suggests that the financial incentives solar power suppliers required to enter the market was high. This might come from high production costs and technology barriers. The sharp jumps in price also implies that when price increased, only small increments of capacity were added, suggesting the idea that the early market had few suppliers and the cost efficiency was quite low.

India transitioned to an e-RA system in 2016. The e-RA system introduced a highly competitive bidding environment, which subsequently reduced electricity prices. The step function we created for 2016 shows a much flatter and smoother supply curve compared to that of 2013 FIT prices, with bid prices concentrated between 2 and 4 Rs./kWh. This suggests that more suppliers were willing to offer larger capacities at lower prices. The phenomenon could be the result of advanced technology and increased market competition. Unlike 2013, where additional capacity required higher tariffs, the 2016 curve exhibits a more gradual price increase, meaning that supply was much more elastic and responsive to lower prices.

Comparing 2013 and 2016, it is evident that the shift to e-RA improved cost efficiency and market competitiveness. The FIT system in 2013 resulted in a less competitive market, where high prices were inevitable to encourage supply, whereas in 2016, the auction-based system led to cost reductions and substantial capacity expansion at lower prices.

5. Policy Outcome Analysis

e-RA mobilized a substantial number of new renewable energy projects in India, but not all of them reached completion on time. Exceptionally successful e-RAs drove the growth of solar energy in India in the mid-2010s (Gulia, Sharma, & Garg, 2023). Repeating successful types of renewable energy tenders followed by transparent e-RAs of highly bankable PPAs underwritten by the Indian government (via the Solar Energy Corporation of India, or SECI) to drive down prices was one of the key components for renewable energy growth in India in the past decade (Gulia, Sharma, & Garg, 2023).

5.1. Issuance, Allotment and Commissioning³

Since 2014, utility-scale renewable energy tenders⁴ have driven India's market growth, with the government expanding its tender portfolio to a range of renewable energy technologies. In 2024, 73 GW of renewable energy of cumulative capacity was issued by tendering agencies, which surpassed historical issuance. The average capacity issued per renewable energy tender has been growing as well, with a 25% increase from 2023 to 2024, which translates to reaching almost 1 GW per tender. Additionally, the issuance share of newer renewable energy technologies in tenders has increased and surpassed 49% in 2024, leading to a greater focus on wind solar hybrids and energy storage systems. This increase was primarily driven by the energy off takers who have begun prioritizing more power output quality (JKM Research, 2025).

While the number of tender issuances has been strongly increasing, particularly since 2023, annual allotted capacity has been growing as well. The latter more than doubled to almost 50 GW from 2023 to 2024. Similarly to issued tenders, allotted tenders switched focus to newer renewable energy technologies, with an increase in their share from 23% in 2023 to 60% in 2024 (JKM Research, 2025).

Using the Institute for Energy Economics and JMK Research and Analytics data, we visualized the issuance and allotment of renewable energy from 2015 to 2024 in the figure below. The visualization shows the volatility in the gap between these two metrics. Issuance has increased, reaching 73 GW in 2024, while allotment lags at 50 GW. The dashed line represents the government's plan to add 50 GW of capacity annually for the next few years to reach the 2030 target of 500 GW in total renewable energy (Press Information Bureau, 2023). While allotment reaches the goal in 2024, it is visible that the gap between issuance and allotment has been increasing post-2021. This gap could have been increasing due to financial constraints and cancellations, among many others.

³ Issuance, allotment, and commissioning refer to the number of tenders announced by the government, the capacity allocated to the winning bidders, and the capacity that is ultimately commissioned.

⁴ The term "tenders" refers to the overall procurement process, where the reverse auction is the core price-setting stage.

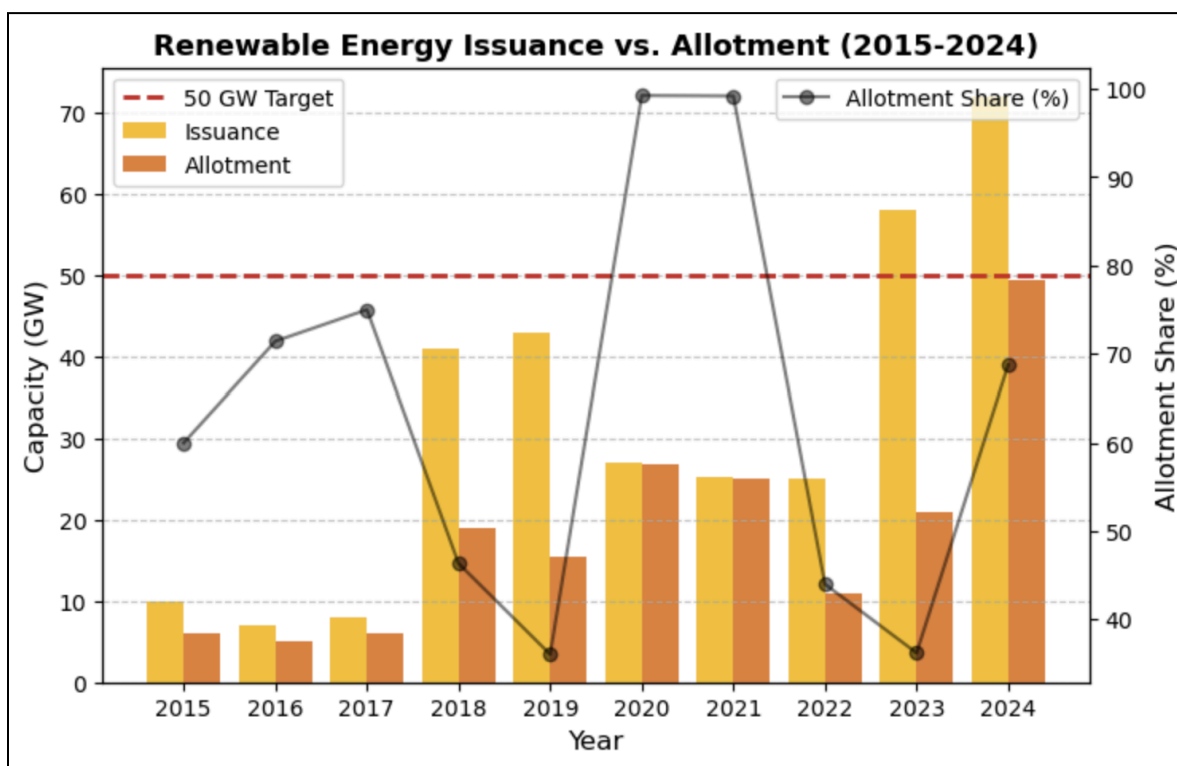


Figure 5: Renewable Energy Issuance vs. Allotment

Source: Visualized using data from Institute for Energy Economics and Financial Analysis and JMK Research & Analytics (2025)

A large portion of auction-rewarded capacity is underdeveloped. For example, a study conducted by Mercom (2023) found that of total solar auctions in 2017, only 58.6% capacity was commissioned by the end of 2018. In another study, the authors computed the total capacity installed as of the target date, December 2022. Only 120.9 GW out of the 175 GW target of renewables were installed, which translates to approximately 69% (Mercom, 2023).

This gap between the rewarded capacity and actual commissioned capacity can often be attributed to the winner's curse, caused by underbidding, which later causes financial challenges, as well as other logistical issues, such as grid collection delays. The previously mentioned Single-Stage Two-Envelope Bid format used in the e-RA aims to address this gap by conducting financial and technical pre-selection, but underperformance remains an issue.

Moreover, many renewable energy tenders get canceled either before or after the auction. Between 2020 and 2024, a total of 38.3 GW capacity was canceled in renewable energy tenders,

32% of which was canceled after the auction. Cancellations occur after the auctions due to under-subscription, which often occurs when the firm is being awarded with less capacity than the capacity tendered or due to delays in signing the power sale agreement. The latter usually takes place when the energy offtakers expect the prices to decline even further, which puts pressure on the firms to bid lower (JKM Research, 2025).

5.2. Winner's Curse

There is evidence indicating the winner's curse in India's 2016 e-RAs in states where developers bid lower than the national average. The developers subsequently faced a high likelihood of failure in deploying solar capacity. This phenomenon occurs when bidders underestimate costs or overestimate profitability. Inevitably, projects that are financially unviable either remain incomplete or operate at a loss.

In Arunachal Pradesh, developers bid 15.4% below the national average price (Rs. 3.5/kWh). However, the installed capacity was a mere 69 kWp, 96.3% lower than the national average (1854.5 kWp). Similar patterns are seen in Assam (-58.8%) and Himachal Pradesh (-78.9%). It might be an indication that low prices were not transferred into successful project deployment. Additionally, while the bid prices in these areas were lower than the national average, the installed capacity was disproportionately lower, which appears counterintuitive to the typical downward-sloping demand curve.

Conversely, Karnataka, where the bid price was slightly above the national average (Rs. 3.839/kWh, +9.7%), had an installed capacity that was 313.2% above the national average. This contrast suggests that developers who bid closer to or above the national average had a higher probability of successful implementation.

2016 India Solar E-Reverse Auction: Bid Prices and Project Deployment Outcomes

Year	State/Region	Lowest Bid Price (Rs./kWh)	National Avg. Bid Price (Rs./kWh)	Deviation from Avg. Price (%)	Total Installed Capacity (kWp)	National Avg. Capacity (kWp)	Deviation from Avg. Capacity (%)	Potential Failure Risk
2016	Arunachal Pradesh	2.96	3.5	-15.4%	69.0	1854.5	-96.3%	High
2016	Assam	2.42	3.5	-30.9%	764.1	1854.5	-58.8%	High
2016	Himachal Pradesh	2.38	3.5	-31.9%	391.2	1854.5	-78.9%	High
2016	Karnataka (for comparison)	3.839	3.5	+9.7%	7660.2	1854.5	+313.2%	Low

This case of adverse selection, has led to a decline in developer participation and increased project delays and cancellations (Gulia, Sharma, & Garg, 2023). In 2012, GoI had to fine 14 PV developers and warn another 14 for being unable to meet their commissioned installation deadlines (del Río & Linares, 2014).

5.3. Prices

Solar module prices decreased considerably by 78% in India and 63% abroad on the global market between 2023 and 2024 (JKM Research, 2025). However, despite the sharp decline, the cost of panel manufacturing in India remained more than double compared to the price on global markets in December 2024 (US\$20.2 per Watt-peak (Wp) in India vs. US\$8.5/Wp globally) (InfoLink, 2024). This shows that although India made notable progress in reducing the cost of solar panel production, it is still behind other countries in efficiency and economies of scale, which would drive down the costs.

Initially upon implementation, the e-RA system successfully reduced domestic tariffs. In 2010, the lowest winning tariff for solar energy was Rs. 10.95/kWh (US\$0.13/kWh), and this number dropped to Rs. 1.99/kWh (US\$0.024/kWh) in the end of 2020 (Gulia, Sharma, & Garg, 2023), which is a 80% decline in tariffs. Compared to the 2009 tariff from the baseline FIT system, in 2010, the tariff achieved through the auction system also dropped by up to 58% (Shrimali, Konda, Farooquee, & Nelson, 2015). However, in recent years, the solar reverse

auction tariffs remained approximately unchanged, ranging between US¢2.5 and US¢ 2.7 per kilowatt hour (kWh) between 2023 and 2024. This is surprising because lower solar module prices during this period would be expected to be reflected in lower tariffs, which is not the case here. The figure below shows that the lowest solar tariffs show stagnation from January 2023 until December 2024, unaffected by the decline in module prices. It shows the lowest winning bids, which means that the overall market price is probably higher.

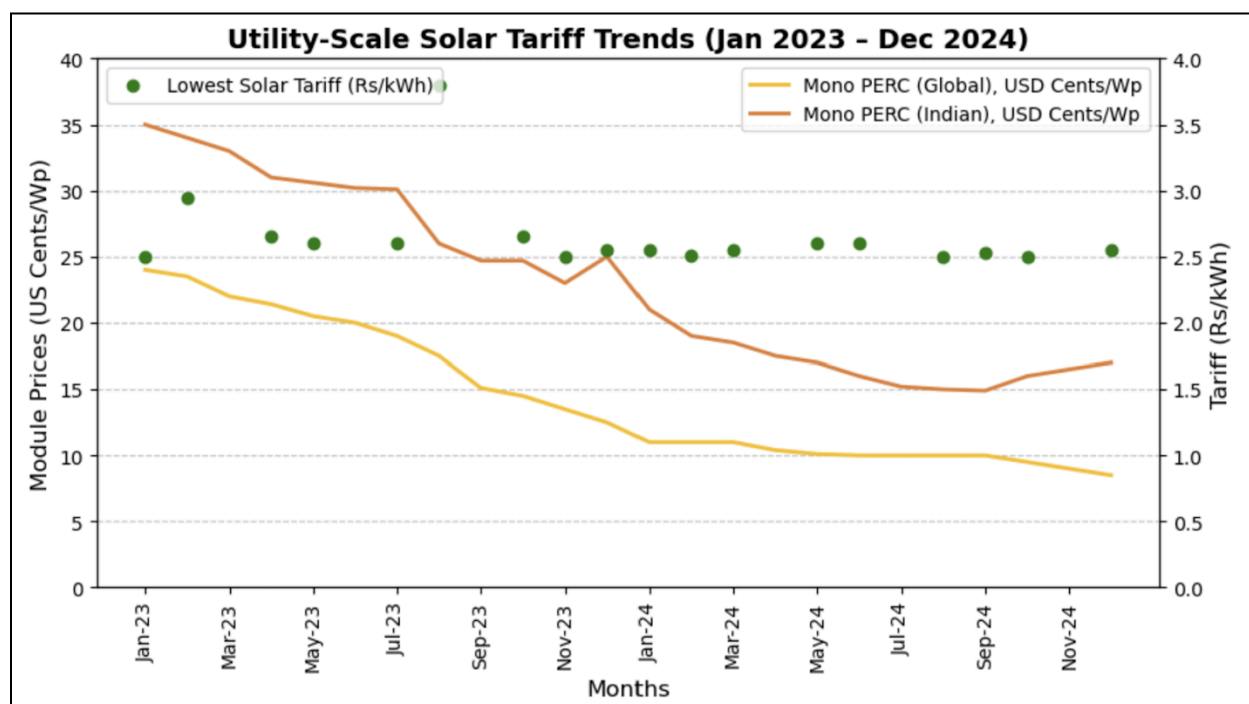


Figure 6: Utility-Scale Solar Tariff Trends

Source: Visualized using data from Institute for Energy Economics and Financial Analysis and JMK Research & Analytics (2025)

One explanation could be that the lowest tariffs are relatively inelastic because the projects need to remain financially viable, and aggressive underbidding may pose a risk to profitability. Additionally, other cost factors, such as land or grid costs, may not be decreasing in costs as much as the modules, making the investors cautious in bidding. Another reason for non-declining solar tariffs is Indian rupee depreciation against the US dollar and limited

availability of the solar modules until 2024, which caused procurement delays and helped sustain high tariffs (JKM Research, 2025).

6. Conclusion

Our analysis highlights gaps in policy implementation, but also notices the drastic positive impact it brings to the renewable energy sector in India. The auction system has been a catalyst to renewable energy growth through driven down prices, and enhancing transparency. However, on the downside, it has introduced challenges like aggressive bidding, lowered participation and project delays & cancellations. To address these challenges, GoI needs to ensure that bidders establish a balance between cost reduction and financial viability, which the current technological and financial bid system is unable to take into account. A more robust evaluation to reveal the true quality of the bidder could address these problems. Additionally, changes such as replacing a maximum tariff price with a minimum floor reserve price as in the LoLA auctions can encourage participation and incentivize bidders to report their true costs (Lopomo, Persico, & Villa, 2022).

Lastly, we believe that the end goal of this policy is to also increase consumption of solar energy. As seen in Fig 1 and 2, there is a dissonance in capacity increase and utilization of renewable energy, something that these policies don't address. The lack of qualified bidders may point to a broader issue concerning the demand for solar power, highlighting the need for further research into this matter.

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